

4.4 Coordinate Systems

I

An important reason for specifying a basis β for a vector space V is to impose a "coordinate system" on V . This section will show that if β contains n vectors, then the coordinate system will make V act like $\mathbb{R}^n$. If V already is $\mathbb{R}^n$ itself, then β will determine a coordinate system that gives a new "view" of V .

The existence of coordinate systems rests on the following fundamental result.

Theorem 8 - The Unique Representation Theorem

Let $\beta = \{\vec{b}_1, \dots, \vec{b}_n\}$ be a basis for a vector space V . Then for each $\vec{x}$ in V , there exists a unique set of scalars $c_1, \dots, c_n$ such that

$$\vec{x} = c_1 \vec{b}_1 + \dots + c_n \vec{b}_n \quad (1)$$

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Proof

Since β spans V , there exist scalars such that (1) holds. Suppose $\vec{x}$ also has the representation

$$\vec{x} = d_1 \vec{b}_1 + \dots + d_n \vec{b}_n$$

for scalars $d_1, \dots, d_n$. Then, subtracting we have

$$\vec{0} = \vec{x} - \vec{x} = (c_1 - d_1) \vec{b}_1 + \dots + (c_n - d_n) \vec{b}_n \quad (2)$$

Since β is linearly independent, the weights in (2) must all be zero. That is $c_j = d_j$ for $1 \leq j \leq n$.

4.4 Coordinate Systems (cont'd)

Definition

Suppose $\beta = \{\vec{b}_1, \dots, \vec{b}_n\}$ is a basis for a vector space V and $\vec{x}$ is in V . The coordinates of $\vec{x}$ relative to the basis β (or the β coordinates of $\vec{x}$) are the weights $c_1, \dots, c_n$ such that $\vec{x} = c_1\vec{b}_1 + \dots + c_n\vec{b}_n$.

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If $c_1, \dots, c_n$ are the β coordinates of $\vec{x}$, then the vector in $\mathbb{R}^n$

$$[\vec{x}]_{\beta} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

is the coordinate vector of $\vec{x}$ (relative to β), or the β -coordinate vector of $\vec{x}$. The mapping $\vec{x} \mapsto [\vec{x}]_{\beta}$ is the coordinate mapping (determined by β).

Example 1

Consider a basis $\beta = \{\vec{b}_1, \vec{b}_2\}$ for $\mathbb{R}^2$, where $\vec{b}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{b}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Suppose $\vec{x}$ in $\mathbb{R}^2$ has the coordinate vector $[\vec{x}]_{\beta} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$. Find $\vec{x}$.

Solution

The β -coordinates of $\vec{x}$ tells us how to build

$\vec{x}$ from the vectors in β . That is,

$$\vec{x} = -2\vec{b}_1 + 3\vec{b}_2 = (-2)\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

Example 2

The entries in the vector $\vec{x} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ are the coordinates of $\vec{x}$ relative to the standard basis $\mathcal{E} = \{\vec{e}_1, \vec{e}_2\}$, since

$$\begin{bmatrix} 1 \\ 6 \end{bmatrix} = 1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 6\begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1\vec{e}_1 + 6\vec{e}_2$$

If $\mathcal{E} = \{\vec{e}_1, \vec{e}_2\}$, then $[\vec{x}]_{\mathcal{E}} = \vec{x}$

A Graphical Interpretation of Coordinates

A coordinate system on a set consists of a one-to-one mapping of the points in the set into $\mathbb{R}^n$. For example, ordinary graph paper provides a coordinate system for the plane when one selects perpendicular axes and a unit of measurement on each axis.

Figure 1 shows the standard basis $\{\vec{e}_1, \vec{e}_2\}$, the vectors $\vec{b}_1 (= \vec{e}_1)$ and $\vec{b}_2$ from Example 1, and the vector $\vec{x} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$.

4.4 Coordinate systems (cont'd)

Figure 2 shows the vectors $\vec{b}_1$, $\vec{b}_2$ and $\vec{x}$ from Figure 1. However, the standard grid is replaced by a grid especially adapted to the basis β in Example 1. The coordinate vector $[\vec{x}]_\beta = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ gives the location of $\vec{x}$ on this new coordinate system: $-2\vec{b}_1 + 3\vec{b}_2$.

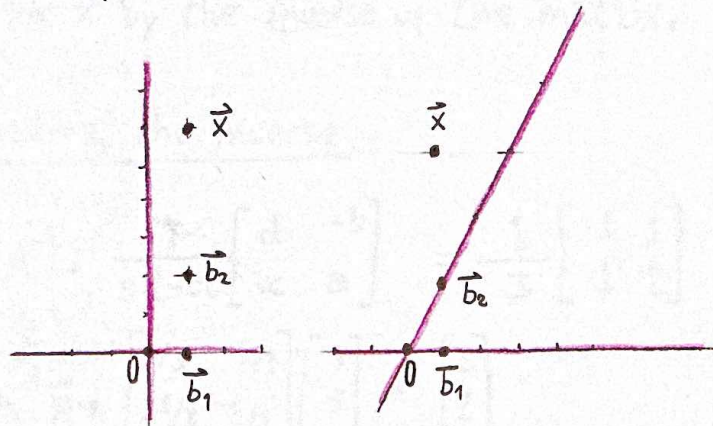


Figure 1

Figure 2

Example 3

In crystallography, the description of a crystal lattice is aided by choosing a basis $\{\vec{u}, \vec{v}, \vec{w}\}$ for $\mathbb{R}^3$ that corresponds to three adjacent edges of one "unit cell" of the crystal. An entire lattice is constructed by stacking together many copies of one cell. There are fourteen basic types of unit cells; three are shown in Figure 3.

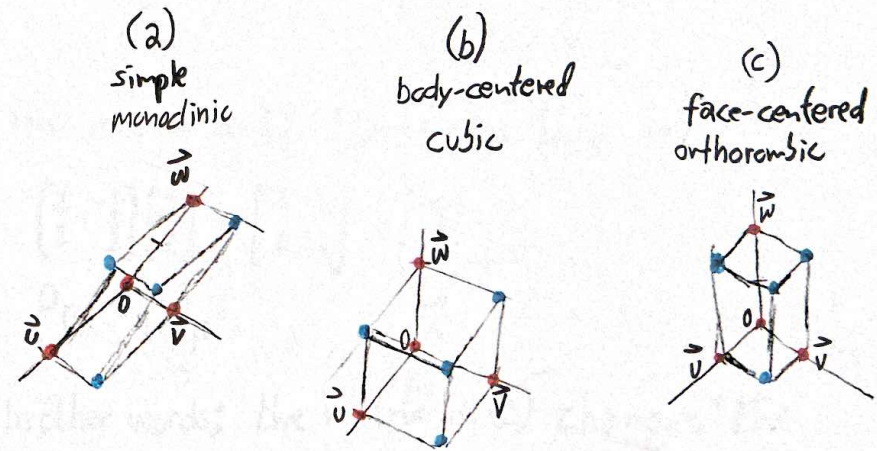


Figure 3- examples of unit cells

The coordinates of atoms within the crystals are given relative to the basis for the lattice.

coordinates in $\mathbb{R}^n$

Example 4

Let $\vec{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\vec{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, and $\beta = \{\vec{b}_1, \vec{b}_2\}$.

Find the coordinate vector $[\vec{x}]_\beta$ of $\vec{x}$ relative to β .

Solution

The β -coordinates of $\vec{x}$, c_1 and c_2 , satisfy

$$c_1 \vec{b}_1 + c_2 \vec{b}_2 = \vec{x}$$

(β is the basis for a ^{vector} subspace)

4.4 Coordinate Systems (cont'd)

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$$\text{or } \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \quad (3)$$

This equation can be solved by row operations on an augmented matrix, or by multiplying the vector $\vec{x}$ by the inverse of the matrix.

Finding the inverse

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{bmatrix}$$

$$A^{-1} \vec{x} = \begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

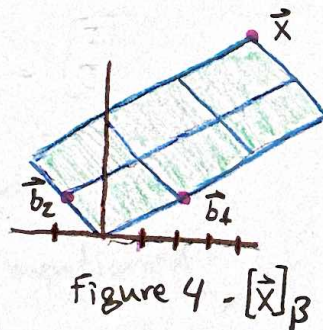
$$[\vec{x}]_{\beta} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Row operations

$$\left[\begin{array}{cc|c} 2 & -1 & 4 \\ 1 & 1 & 5 \end{array} \right] \begin{array}{l} r_1 \leftrightarrow r_2 \\ r_2 \leftarrow r_2 - 2r_1 \end{array} \sim \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -6 \end{array} \right]$$

$$\begin{array}{l} r_2 \leftarrow r_2 \cdot (-1/3) \\ r_1 \leftarrow r_1 - r_2 \end{array} \sim \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \end{array} \right]$$

$$[\vec{x}]_{\beta} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$



The matrix in (3) transforms $[\vec{x}]_{\beta}$ into $\vec{x}$.

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6-2 \\ 3+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$P_{\beta} \cdot [\vec{x}]_{\beta} = \vec{x}$$

In other words; the matrix in (3) changes the β -coordinates of a vector $\vec{x}$ into the standard coordinates for $\vec{x}$.

An analogous change of coordinates can be carried out in $\mathbb{R}^n$ for a basis $\beta = \{\vec{b}_1, \dots, \vec{b}_n\}$.

Let

$$P_{\beta} = [\vec{b}_1 \ \vec{b}_2 \ \dots \ \vec{b}_n]$$

Then the vector equation

$$\vec{x} = c_1 \vec{b}_1 + c_2 \vec{b}_2 + \dots + c_n \vec{b}_n$$

is equivalent to

$$\vec{x} = P_{\beta} [\vec{x}]_{\beta}$$

(4)

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We call P_β the change-of-coordinate matrix from β to the standard basis in $\mathbb{R}^n$.

Left multiplication by P_β transforms the coordinate vector $[\vec{x}]_\beta$ into $\vec{x}$.

Since the columns of P_β form a basis for $\mathbb{R}^n$, P_β is invertible (by the invertible matrix theorem, pp 145, 189). Thus, left multiplication by P_β^{-1} converts $\vec{x}$ into its β -coordinate vector:

$$P_\beta^{-1} \vec{x} = [\vec{x}]_\beta$$

The correspondence $\vec{x} \mapsto [\vec{x}]_\beta$, produced by P_β^{-1} , is the coordinate mapping mentioned earlier (pp. 91/92).

The Coordinate Mapping

Choosing a basis $\beta = \{\vec{b}_1, \dots, \vec{b}_n\}$ for a vector space V introduces a coordinate system in V . The coordinate mapping $\vec{x} \mapsto [\vec{x}]_\beta$ connects the possibly unfamiliar space V to the familiar space $\mathbb{R}^n$. See Figure 5, points in V can now be identified by new "names".

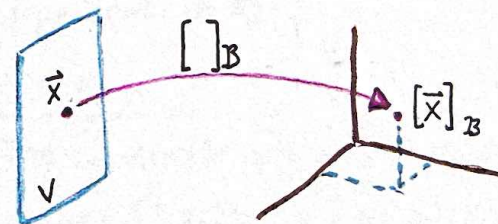


Figure 5 - The coordinate mapping from V onto $\mathbb{R}^n$.

Theorem 9

Let $\beta = \{\vec{b}_1, \dots, \vec{b}_n\}$ be a basis for a vector space V . Then the coordinate mapping $\vec{x} \mapsto [\vec{x}]_\beta$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

4.4 Coordinate Systems (cont'd)

The linearity of the coordinate mapping extends to linear combinations, just as in Section 1.8.

If $\vec{v}_1, \dots, \vec{v}_p$ are in V and if $c_1, \dots, c_p$ are scalars, then

$$[c_1 \vec{v}_1 + \dots + c_p \vec{v}_p]_{\mathcal{B}} = c_1 [\vec{v}_1]_{\mathcal{B}} + \dots + c_p [\vec{v}_p]_{\mathcal{B}} \quad (5)$$

In words, (5) says that the $\mathcal{B}$ -coordinate vector of a linear combination of $\vec{v}_1, \dots, \vec{v}_p$ is the same linear combination of their coordinate vectors.

Example 5

Let $\mathcal{B}$ be the standard basis of the space $\mathbb{P}_3$ of polynomials; that is, let

$$\mathcal{B} = \{1, t, t^2, t^3\}.$$

A typical element $\vec{p}$ of $\mathbb{P}_3$ has the form

$$\vec{p}(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

Since $\vec{p}$ is already displayed as a linear combination of the standard basis vectors, we conclude that

$$[\vec{p}]_{\mathcal{B}} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

Thus the coordinate mapping $\vec{p} \mapsto [\vec{p}]_{\mathcal{B}}$ is an isomorphism from $\mathbb{P}_3$ onto $\mathbb{R}^4$. All vector space operations in $\mathbb{P}_3$ corresponds to operations in $\mathbb{R}^4$.

Example 6

Use coordinate vectors to verify that the polynomials $1 + 2t^2$, $4 + t + 5t^2$, and $3 + 2t$ are linearly dependent in $\mathbb{P}_2$.

Solution

The coordinate mapping from Example 5 produces vectors $(1, 0, 2)$, $(4, 1, 5)$, and $(3, 2, 0)$, respectively. Writing these vectors as columns in a Matrix A , we can check dependency by solving $A\vec{x} = \vec{0}$.

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 2 & 5 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -5 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Thus, the polynomial relations are:

$$2(4 + t + 5t^2) - 5(1 + 2t^2) = 3 + 2t$$

4.4 Coordinate Systems (cont'd)

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The final example concerns a plane in $\mathbb{R}^3$ that is isomorphic to $\mathbb{R}^2$.

Example 7

Let $\vec{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$, and $B = \{\vec{v}_1, \vec{v}_2\}$.

Then B is basis for $H = \text{Span}\{\vec{v}_1, \vec{v}_2\}$. Determine if $\vec{x}$ is in H , and if it is, find the coordinate vector of $\vec{x}$ relative to B .

Solution

By the definition on p.256, if $\vec{x}$ is in H , then there must exist scalars, c_1, c_2 , such that

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

then these scalars, c_1, c_2 , will be our B -coordinates of $\vec{x}$. We can find c_1, c_2 , if they exist, by row operations

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

thus $c_1 = 2$, $c_2 = 3$.

Giving us $[\vec{x}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Also, $\vec{x}$ is in H .

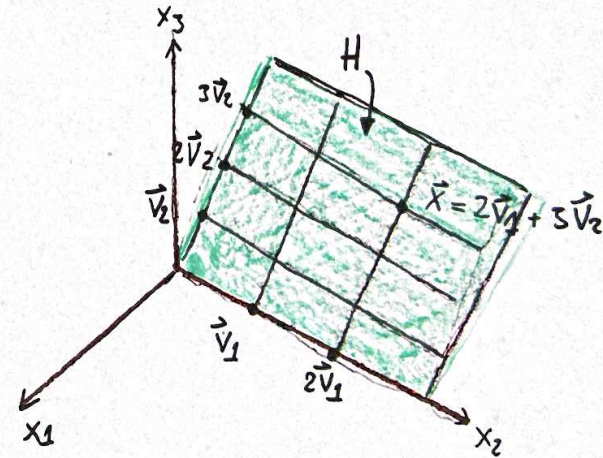


Figure 6 - A coordinate system on a plane H in $\mathbb{R}^3$.